

Readme_for_running_simulation

By: Ghanashyam R K P <ghanashyamrkp@gmail.com>

This document will be a guide to help simulate the video and take further action with the project.

The package to be used for running the simulation is `catkin2_ws` from the drive link sent. This package is the catkin package that is the standard for running ROS programs.

So open a Terminal, Run the following command to open both Rviz and Gazebo for simulation purposes.

- First go to the destined folder by running
 - o `cd catkin2_ws/src`
 - o `catkin_make`
- Then run the next command to bring the arm to the environment and start the welding simulation process (as in the video it is just a representation of the arm following a linear path, with the model of auger-point cloud (just points visible - not in the real environment) rotating.
 - `roslaunch kuka_d435 moveit_gazebo.launch`
- This launch file contains all the commands - which will call rviz and set-up gazebo environment, and also call the program (`move_group_python_node`) in the `kuka_arm_test` package, which is accessible by the path - `catkin2_ws/src/kuka_arm_test` and the script for the algorithm for the arm to follow the welding path is in scripts folder.

Some more information

- The “`Angle_Generator_and_Point-Cloud_UofM_MITACS.ipynb`” notebook is used to generate angles and save the point-cloud, which can be used in the “`move_group_python_interface_tutorial.py`” (Refer the picture below).

Readme_for_running_simulation

By: Ghanashyam R K P <ghanashyamrkp@gmail.com>

```
def RotatePointCloud(self):  
    r = rospy.Rate(0.5)  
  
    print(self.move_group.get_current_rpy())  
  
    with open('/home/ghans/catkin2_ws/src/kuka_arm_test/scripts/Correct_Angles.npy', 'rb') as f:  
        angles = np.load(f)  
  
    int_pcd = o3d.io.read_point_cloud("/home/ghans/catkin2_ws/src/kuka_arm_test/scripts/red2.pcd")  
  
    for i in range(83):  
        arr = np.asarray(int_pcd.points)  
  
        r_header = std_msgs.msg.Header()  
        r_header.stamp = rospy.Time.now()  
        r_header.frame_id = self.planning_frame  
  
        ros_pcd = pcl2.create_cloud_xyz32(r_header, arr.tolist())  
  
        self.pub.publish(ros_pcd)  
  
        self.go_to_pose_goal(arr, i)  
  
        pcd_rotmat = copy.deepcopy(int_pcd)  
        R = pcd_rotmat.get_rotation_matrix_from_axis_angle(np.array([0, -(angles[i]), 0]))  
  
        int_pcd.rotate(R)  
  
        r.sleep()
```

- So in the 4th and 5th Line of the function Rotate_Point_Cloud, you can specify the path name of the point cloud of the welding path that you want to rotate and the angles - numpy array file and save it. After this follow - do a catkin_make and follow the above commands to see the new results. (File is in catkin2_ws/src/kuka_arm_test)

Additional Point:

Readme_for_running_simulation

By: Ghanashyam R K P <ghanashyamrpk@gmail.com>

https://drive.google.com/file/d/1sYNmZ-oXY0lgNGADIOTX_Yb5ocNVgvPg/view?resourcekey - as shown in this video, to make the point-cloud visible import the file “red2.pcd” in the “kuka_arm_test” package into the Rviz using the plugin inbuilt in rviz to see point-cloud rotating.

For any further doubts, shoot an email anytime to the email-id in the header of the pages.

-----Thank You -----